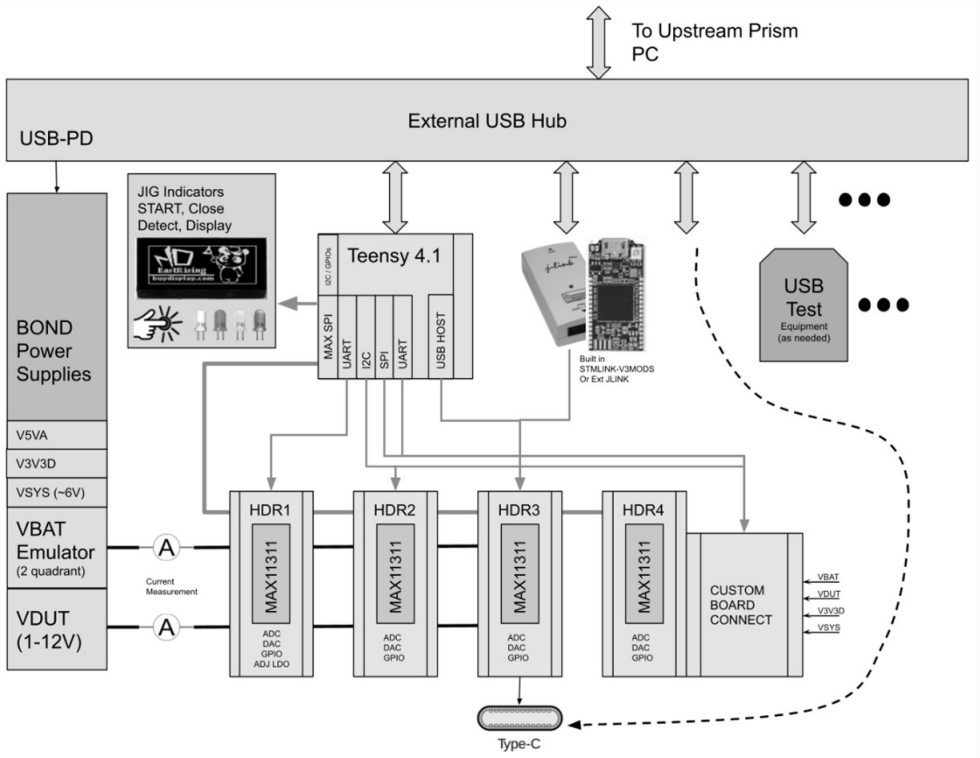


Prism Teensy Board level Test (BLT)

Refer to www.sistemi.ca for Prism/Lente Test Framework.

Testing Capabilities:

Each Header (1, 2, 3, 4) exposes the MAX11311 chip pins to the DUT. Each pin can be configured for either GPIO, DAC (12 bit) or ADC (12 bit), and output range of 0-2.5/5/10V.

In Addition to the MAX11311 each Header provides additional functions,

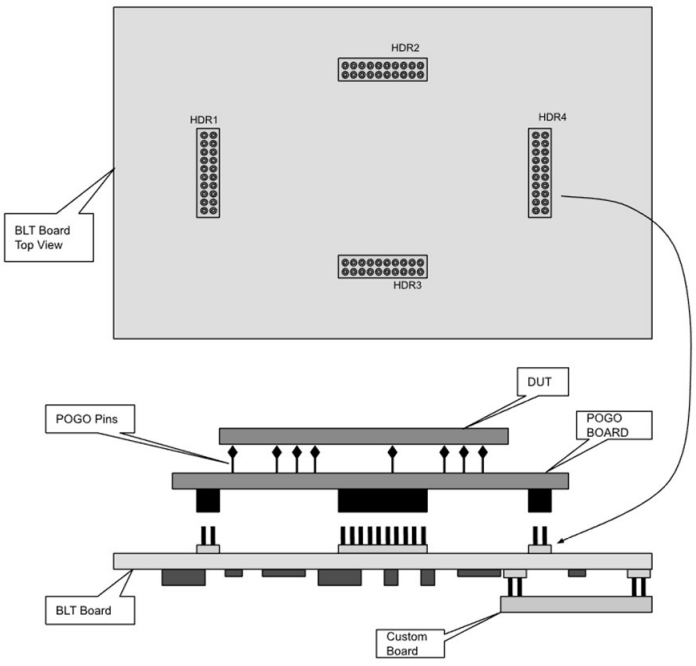
1] VBAT Emulator, sinking and sourcing current, with current measurement, 1.5-5V.
VDUT (2-12V) Supply with current measurement.
Additional Supply 1-5V.
UART (with level shifter)

2] I2C (with level shifter)

3] Segger JTAG (4 wire)
USB Slave to PC or Teensy

4] Connections to expansion Custom PCB
Custom PCB - create specific test measurement or stimulous circuits
- Connections for Power (3,3V, VBAT, VBUS, 6V)
- I2C and UART connections to Teensy

Stack Up



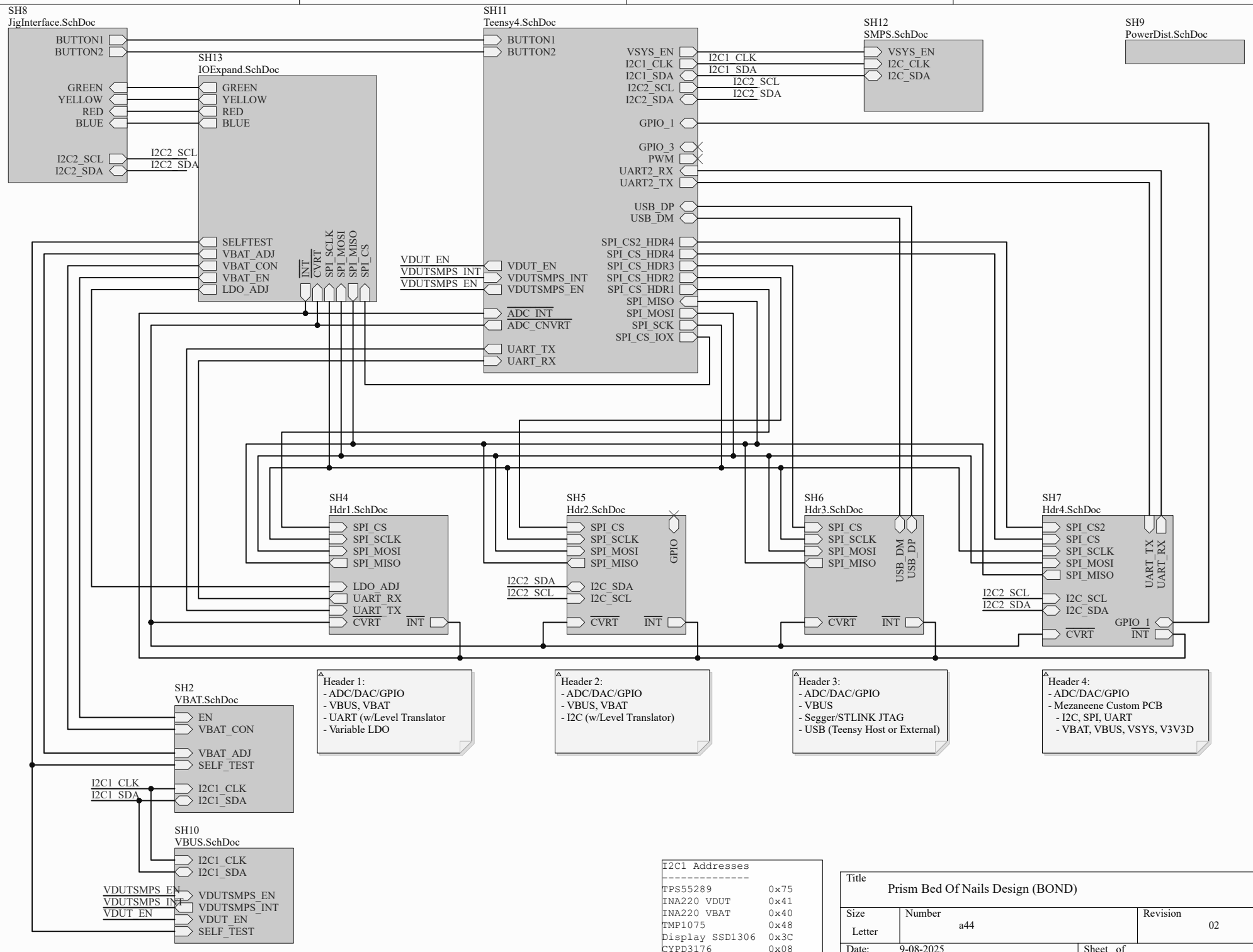
Stack Up Notes

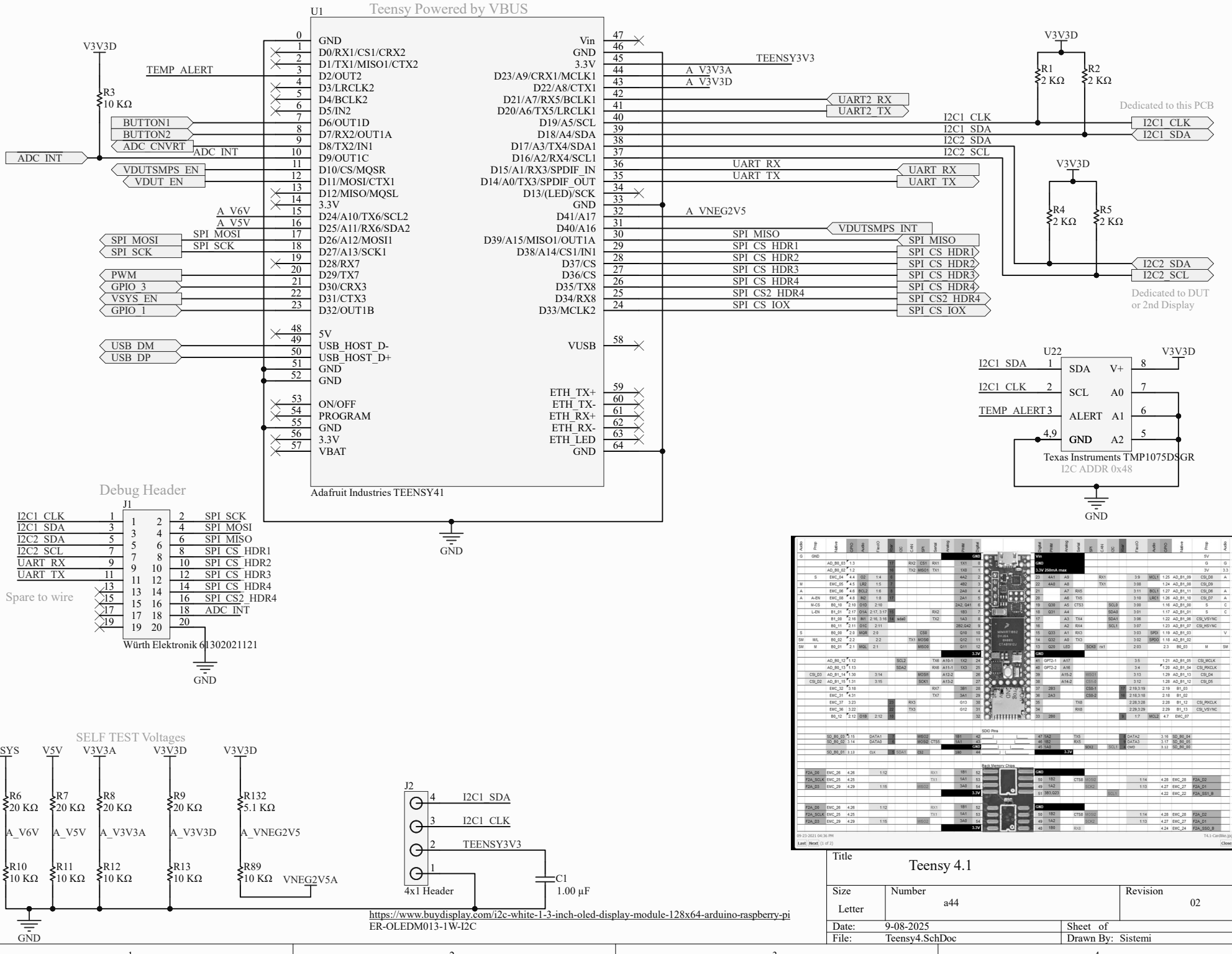
1) The BLT board provides test measurements and stiumlous.

2) The Pogo Board interfaces the BLT board to the Device Under Test (DUT) via spring loaded probes.

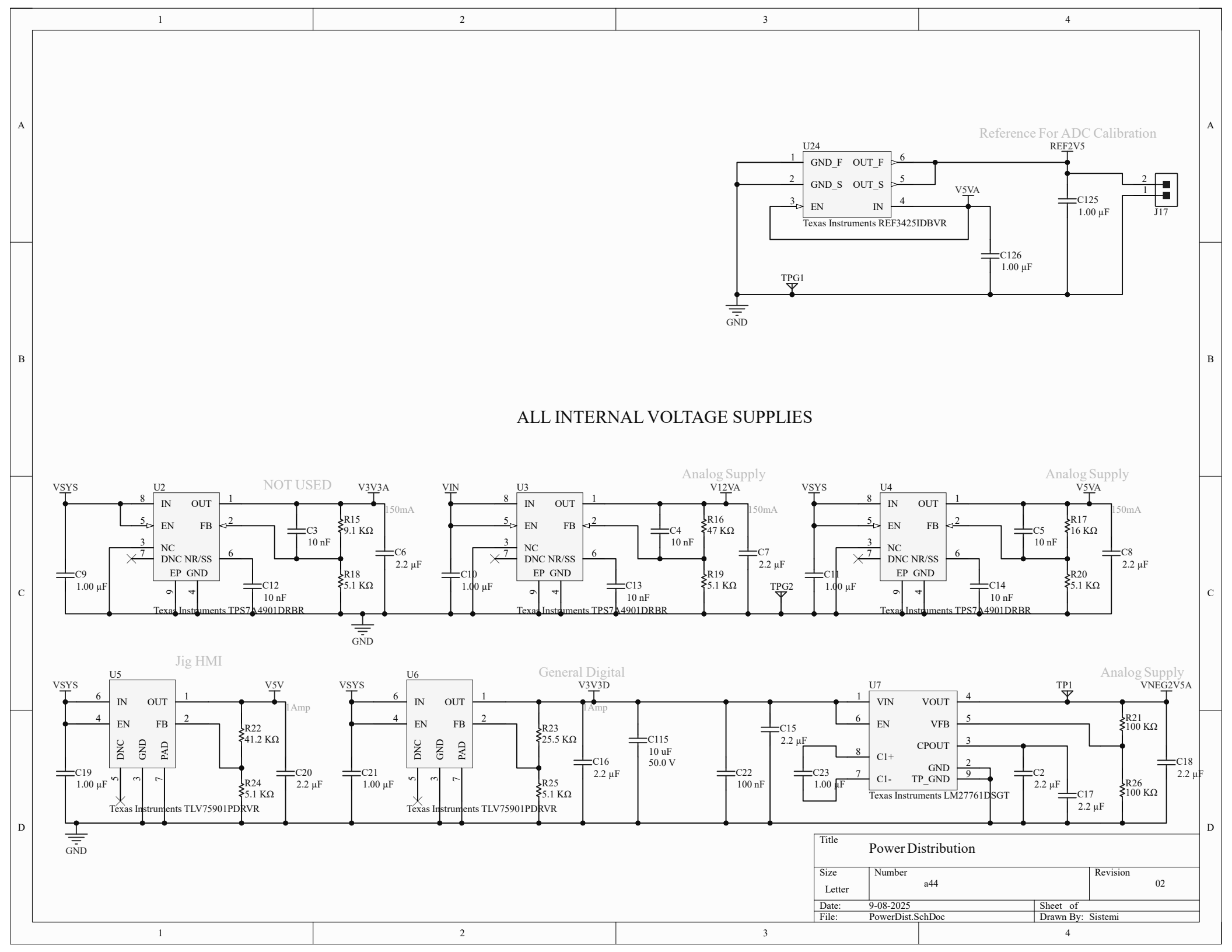
Various DUTs can be supported by one BLT board by using different Pogo Boards.

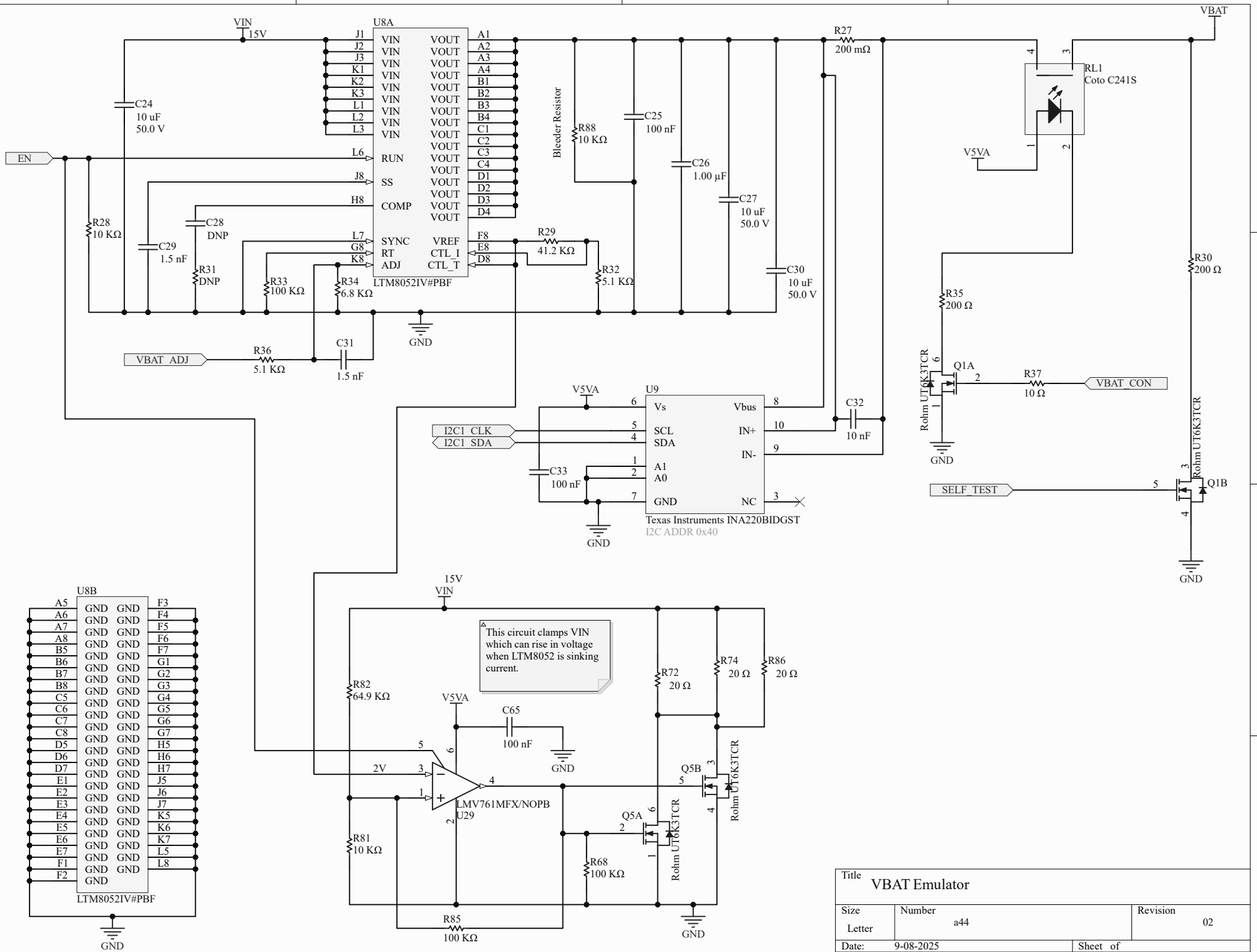
Title: BOND Block Diagram and Notes		
Size: Letter	Number: a44	Revision: 02
Date: 9-08-2025		Sheet of
File: BlockDiag.SchDoc		Drawn By: Sistemi

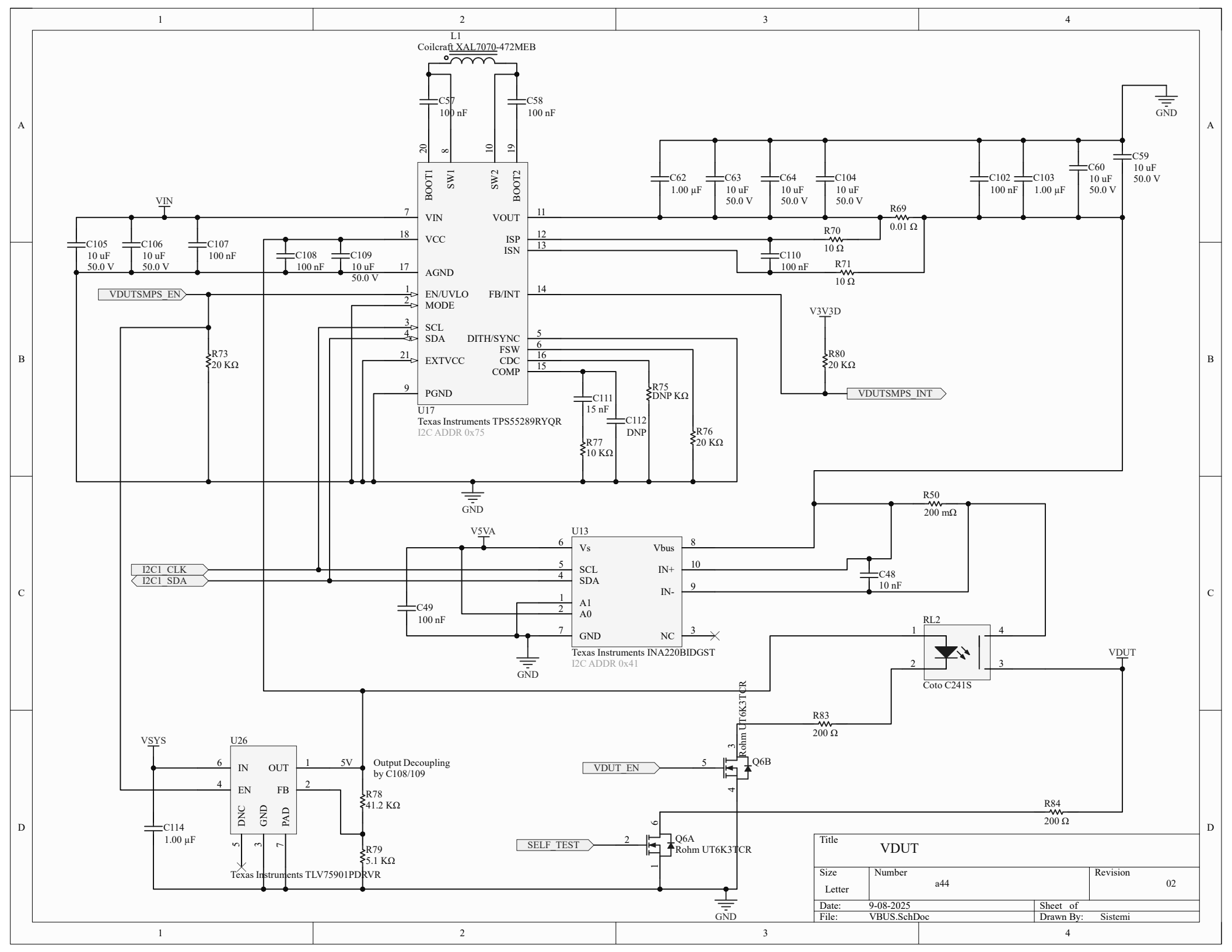

SH8
JigInterface.SchDoc
BUTTON1
BUTTON2
GREEN
YELLOW
RED
BLUE
I2C2_SCL
I2C2_SDA
SH13
IOExpand.SchDoc
SELFTEST
VBAT_ADJ
VBAT_CON
VBAT_EN
LDO_ADJ
INT
CVRT
SPI_SCLK
SPI_MOSI
SPI_MISO
SPI_CS
SH11
Teensy4.SchDoc
VDUT_EN
VDUTSMPS_INT
VDUTSMPS_EN
ADC_INT
ADC_CNVRT
UART_TX
UART_RX
VSYS_EN
I2C1_CLK
I2C1_SDA
I2C2_SCL
I2C2_SDA
GPIO_1
GPIO_3
PWM
UART2_RX
UART2_TX
USB_DP
USB_DM
SPI_CS2_HDR4
SPI_CS_HDR4
SPI_CS_HDR3
SPI_CS_HDR2
SPI_CS_HDR1
SPI_MISO
SPI_MOSI
SPI_SCK
SPI_CS_IOX
SH12
SMPS.SchDoc
SH9
PowerDist.SchDoc
SH4
Hdr1.SchDoc
SH5
Hdr2.SchDoc
GPIO
SH6
Hdr3.SchDoc
USB_DM
USB_DP
SH7
Hdr4.SchDoc
SPI_CS2
UART_TX
UART_RX
GPIO_1
Header 1:
- ADC/DAC/GPIO
- VBUS, VBAT
- UART (w/Level Translator
- Variable LDO
Header 2:
- ADC/DAC/GPIO
- VBUS, VBAT
- I2C (w/Level Translator)
Header 3:
- ADC/DAC/GPIO
- VBUS
- Segger/STLINK JTAG
- USB (Teensy Host or External)
Header 4:
- ADC/DAC/GPIO
- Mezaneene Custom PCB
- I2C, SPI, UART
- VBAT, VBUS, VSYS, V3V3D
SH2
VBAT.SchDoc
EN
VBAT_CON
VBAT_ADJ
SELF_TEST
SH10
VBUS.SchDoc
VDUTSMPS_EN
VDUTSMPS_INT
VDUT_EN
SELF_TEST
I2C1 Addresses
TPS55289 0x75
INA220 VDUT 0x41
INA220 VBAT 0x40
TMP1075 0x48
Display SSD1306 0x3C
CYPD3176 0x08
Title
Prism Bed Of Nails Design (BOND)
Size
Letter
Number
a44
Revision
02
Date: 9-08-2025
Sheet of
File: Top.SchDoc
Drawn By: Sistemi

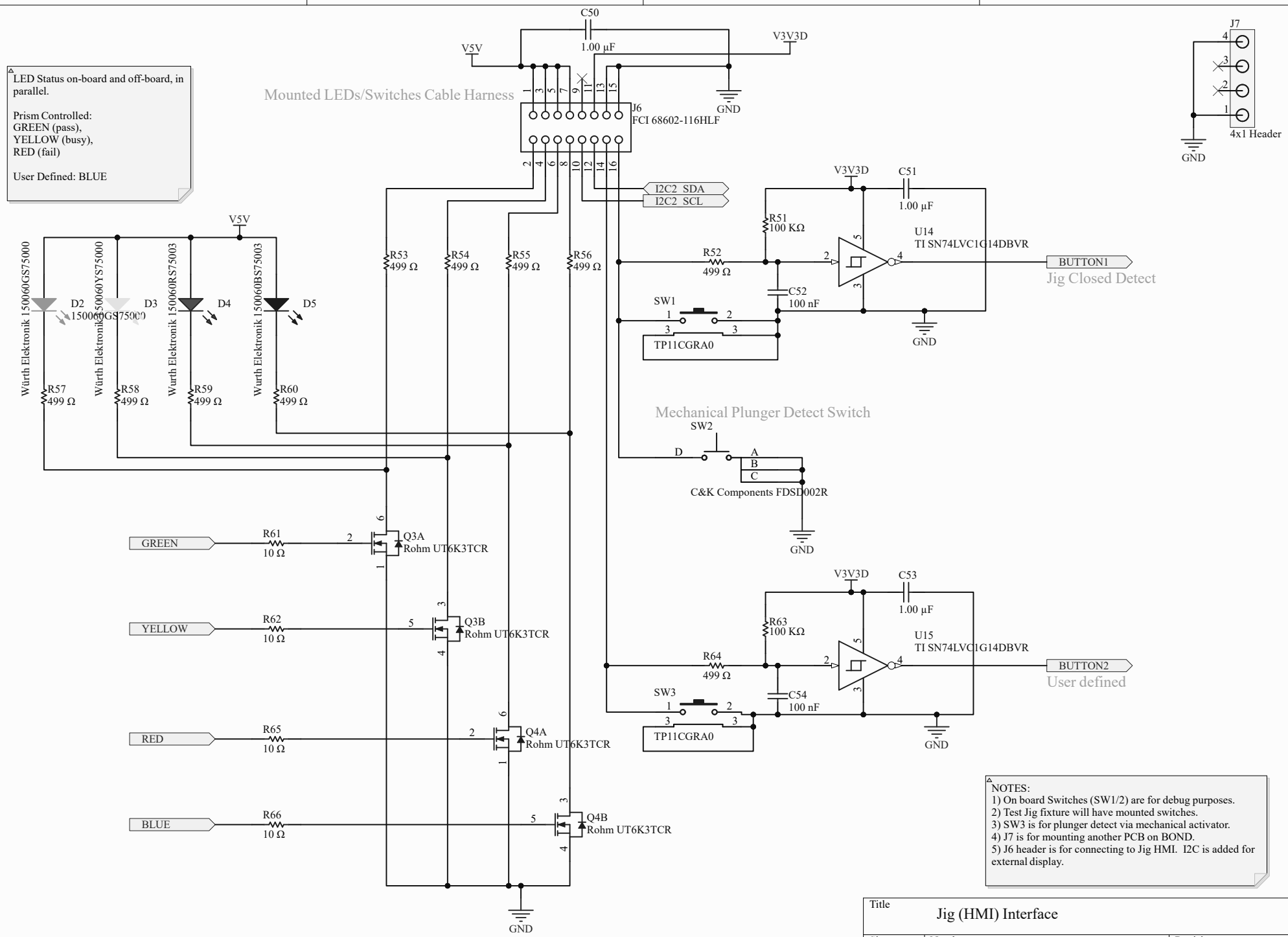


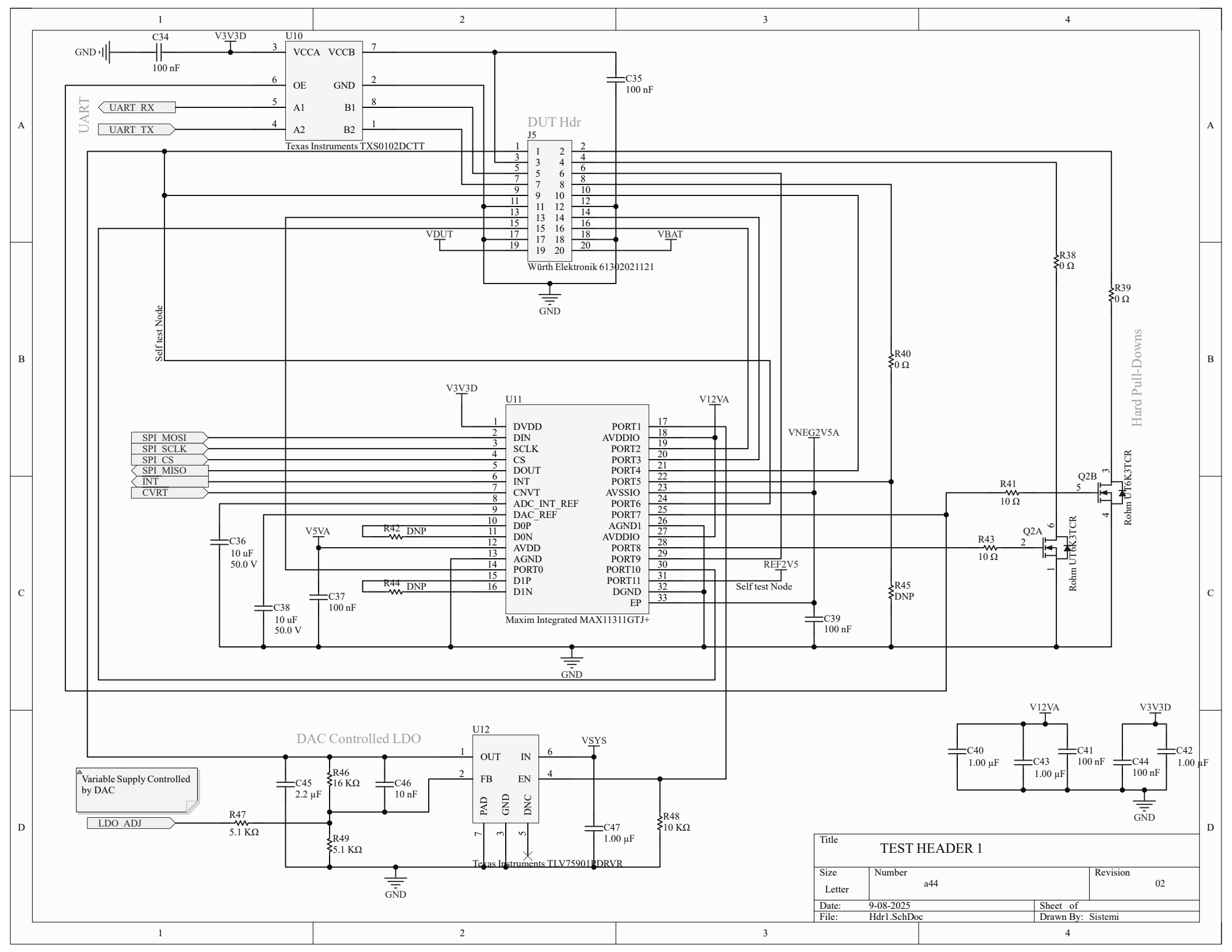
Title	Teensy 4.1		
Size: Letter	Number: a44	Revision: 02	
Date: 9-08-2025		Sheet of	
File: Teensy4.SchDoc		Drawn By: Sistemi	

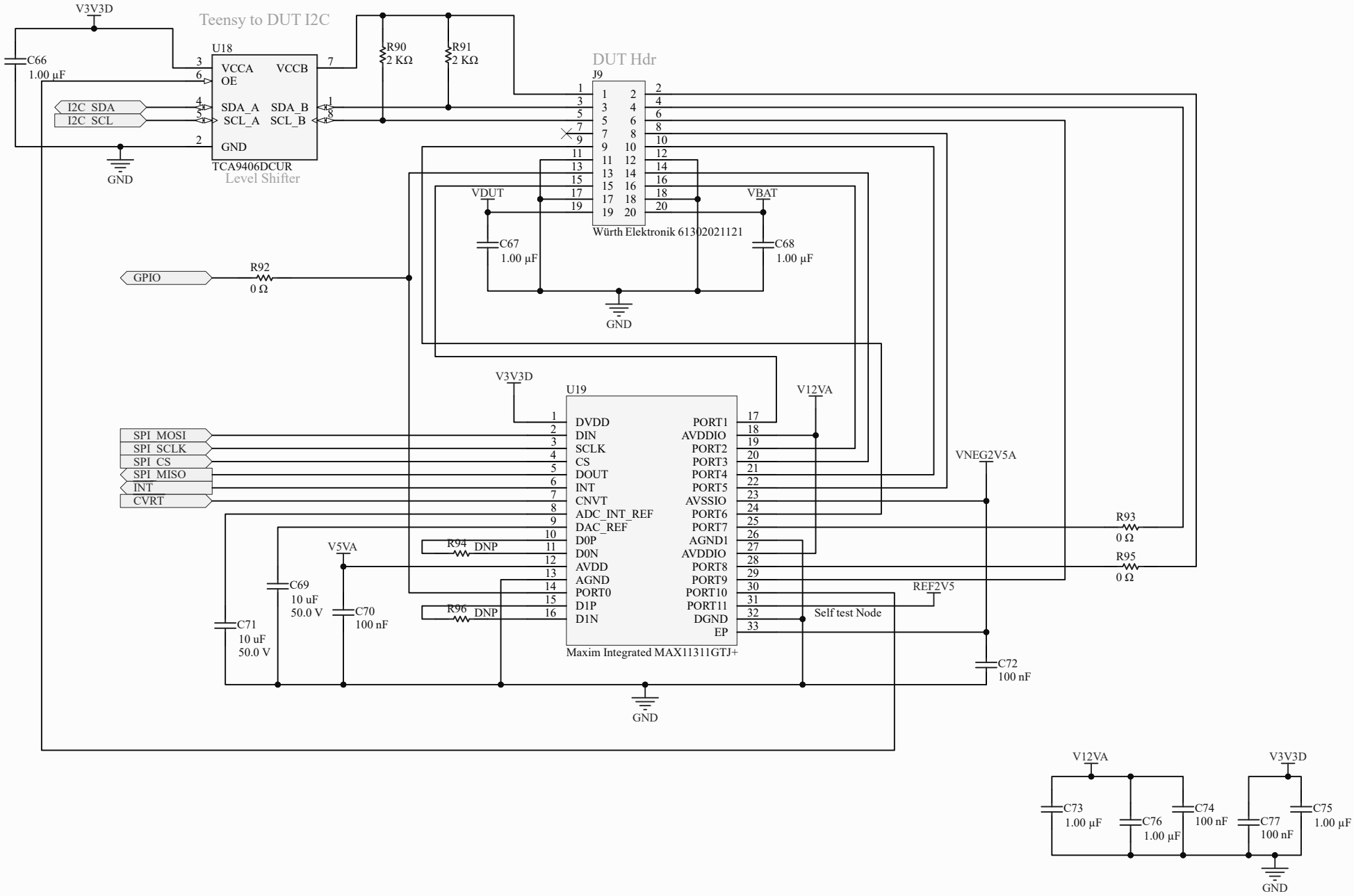

Reference For ADC Calibration
U24
GND_F
OUT_F
GND_S
OUT_S
EN
IN
Texas Instruments REF3425IDBVR
REF2V5
V5VA
C125
1.00 µF
J17
C126
1.00 µF
TPG1
GND
ALL INTERNAL VOLTAGE SUPPLIES
NOT USED
U2
VSYS
V3V3A
150mA
C9
1.00 µF
C3
10 nF
R15
9.1 KΩ
C6
2.2 µF
C12
10 nF
R18
5.1 KΩ
Texas Instruments TPS7A4901DRBR
Analog Supply
U3
VIN
V12VA
C10
1.00 µF
C4
10 nF
R16
47 KΩ
C7
2.2 µF
C13
10 nF
R19
5.1 KΩ
TPG2
Analog Supply
U4
VSYS
V5VA
C11
1.00 µF
C5
10 nF
R17
16 KΩ
C8
2.2 µF
C14
10 nF
R20
5.1 KΩ
Jig HMI
U5
VSYS
V5V
1Amp
C19
1.00 µF
R22
41.2 KΩ
C20
2.2 µF
R24
5.1 KΩ
Texas Instruments TLV75901PDRVR
General Digital
U6
VSYS
V3V3D
1Amp
C21
1.00 µF
R23
25.5 KΩ
C16
2.2 µF
C115
10 uF
50.0 V
R25
5.1 KΩ
Texas Instruments TLV75901PDRVR
C22
100 nF
C15
2.2 µF
C23
1.00 µF
U7
VIN
VOUT
EN
VFB
CPOUT
C1+
GND
C1-
TP_GND
Texas Instruments LM27761DSGT
TP1
Analog Supply
VNEG2V5A
R21
100 KΩ
C18
2.2 µF
C2
2.2 µF
C17
2.2 µF
R26
100 KΩ
GND
Title
Power Distribution
Size
Letter
Number
a44
Revision
02
Date:
9-08-2025
File:
PowerDist.SchDoc
Sheet of
Drawn By:
Sistemi


VIN
15V
U8A
LTM8052IV#PBF
EN
C24
10 uF
50.0 V
R28
10 KΩ
C29
1.5 nF
C28
DNP
R31
DNP
R33
100 KΩ
R34
6.8 KΩ
R88
10 KΩ
Bleeder Resistor
C25
100 nF
C26
1.00 µF
C27
10 uF
50.0 V
C30
10 uF
50.0 V
R27
200 mΩ
R29
41.2 KΩ
R32
5.1 KΩ
VBAT_ADJ
R36
5.1 KΩ
C31
1.5 nF
GND
VBAT
RL1
Coto C241S
V5VA
R35
200 Ω
R30
200 Ω
Q1A
Rohm UT6K3TCR
R37
10 Ω
VBAT_CON
SELF_TEST
Q1B
U9
Texas Instruments INA220BIDGST
I2C ADDR 0x40
I2C1_CLK
I2C1_SDA
C32
10 nF
C33
100 nF
U8B
LTM8052IV#PBF
15V
VIN
This circuit clamps VIN which can rise in voltage when LTM8052 is sinking current.
R82
64.9 KΩ
R81
10 KΩ
R85
100 KΩ
R68
100 KΩ
C65
100 nF
2V
LMV761MFX/NOPB
U29
Q5A
Q5B
R72
20 Ω
R74
20 Ω
R86
20 Ω
Title
VBAT Emulator
Size
Letter
Number
a44
Revision
02
Date:
9-08-2025
File:
VBAT.SchDoc
Sheet of
Drawn By:
Sistemi

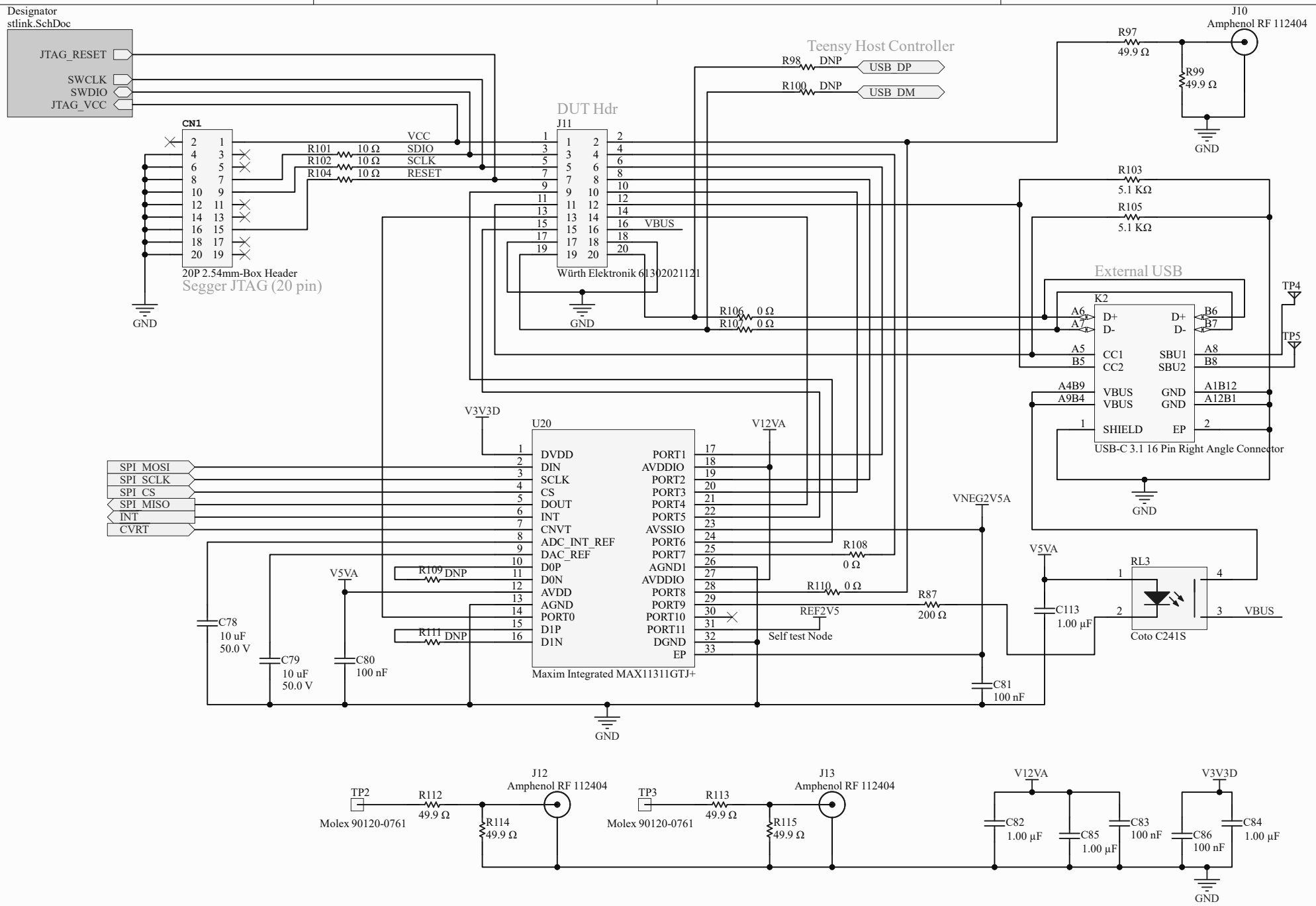

L1
Coilcraft XAL7070-472MEB
C57
100 nF
C58
100 nF
BOOT1
SW1
SW2
BOOT2
VIN
VOUT
VCC
ISP
ISN
AGND
EN/UVLO
MODE
FB/INT
SCL
SDA
DITH/SYNC
FSW
CDC
COMP
EXTVCC
PGND
U17
Texas Instruments TPS55289RYQR
I2C ADDR 0x75
C62
1.00 µF
C63
10 uF
50.0 V
C64
10 uF
50.0 V
C104
10 uF
50.0 V
C102
100 nF
C103
1.00 µF
C60
10 uF
50.0 V
C59
10 uF
50.0 V
GND
R69
0.01 Ω
R70
10 Ω
R71
10 Ω
C110
100 nF
C105
10 uF
50.0 V
C106
10 uF
50.0 V
C107
100 nF
C108
100 nF
C109
10 uF
50.0 V
VDUTSMPS_EN
R73
20 KΩ
V3V3D
R80
20 KΩ
VDUTSMPS_INT
C111
15 nF
C112
DNP
R75
DNP KΩ
R76
20 KΩ
R77
10 KΩ
R50
200 mΩ
V5VA
U13
Vs
Vbus
SCL
SDA
IN+
IN-
A1
A0
GND
NC
Texas Instruments INA220BIDGST
I2C ADDR 0x41
I2C1_CLK
I2C1_SDA
C48
10 nF
C49
100 nF
RL2
Coto C241S
VDUT
R83
200 Ω
R84
200 Ω
Rohm UT6K3TCR
Q6B
VDUT_EN
Q6A
Rohm UT6K3TCR
SELF_TEST
VSYS
U26
IN
OUT
EN
FB
DNC
GND
PAD
5V
Output Decoupling
by C108/109
R78
41.2 KΩ
R79
5.1 KΩ
C114
1.00 µF
Texas Instruments TLV75901PDRVR
Title
VDUT
Size
Letter
Number
a44
Revision
02
Date:
9-08-2025
File:
VBUS.SchDoc
Sheet of
Drawn By:
Sistemi


LED Status on-board and off-board, in parallel.
Prism Controlled:
GREEN (pass),
YELLOW (busy),
RED (fail)
User Defined: BLUE
Mounted LEDs/Switches Cable Harness
C50
1.00 µF
V5V
V3V3D
GND
J6
FCI 68602-116HLF
J7
4x1 Header
I2C2_SDA
I2C2_SCL
D2
150060GS75000
D3
150060YS75000
D4
150060RS75003
D5
150060BS75003
Würth Elektronik
R53 499 Ω
R54 499 Ω
R55 499 Ω
R56 499 Ω
R57 499 Ω
R58 499 Ω
R59 499 Ω
R60 499 Ω
R51 100 KΩ
R52 499 Ω
C51 1.00 µF
U14
TI SN74LVC1G14DBVR
BUTTON1
Jig Closed Detect
SW1
TP11CGRA0
C52 100 nF
Mechanical Plunger Detect Switch
SW2
C&K Components FDSD002R
GREEN
YELLOW
RED
BLUE
R61 10 Ω
R62 10 Ω
R65 10 Ω
R66 10 Ω
Q3A
Q3B
Q4A
Q4B
Rohm UT6K3TCR
R63 100 KΩ
R64 499 Ω
C53 1.00 µF
U15
TI SN74LVC1G14DBVR
BUTTON2
User defined
SW3
TP11CGRA0
C54 100 nF
NOTES:
1) On board Switches (SW1/2) are for debug purposes.
2) Test Jig fixture will have mounted switches.
3) SW3 is for plunger detect via mechanical activator.
4) J7 is for mounting another PCB on BOND.
5) J6 header is for connecting to Jig HMI. I2C is added for external display.
Title
Jig (HMI) Interface
Size
Letter
Number
a44
Revision
02
Date:
9-08-2025
Sheet of
File:
JigInterface.SchDoc
Drawn By:
Sistemi

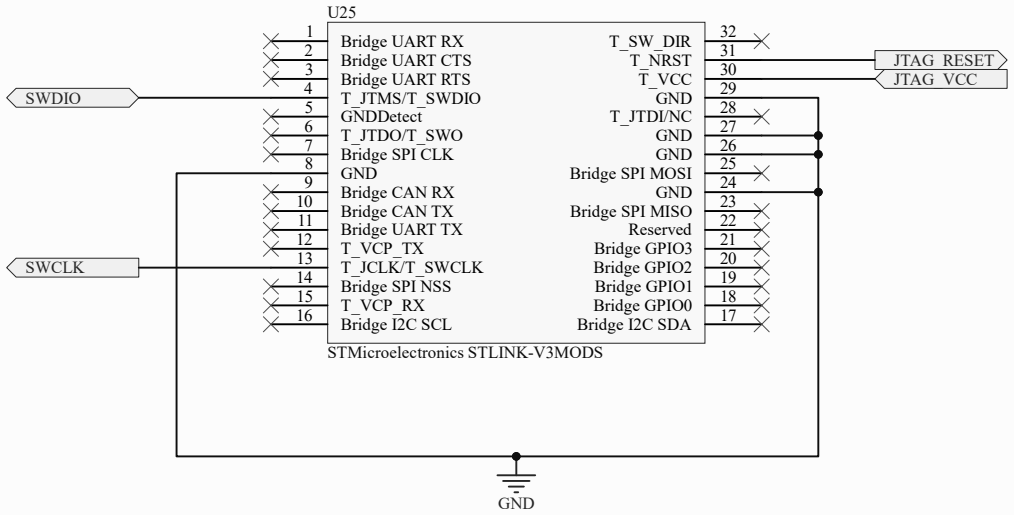

UART
UART_RX
UART_TX
U10
VCCA
VCCB
OE
GND
A1
B1
A2
B2
Texas Instruments TXS0102DCTT
C34
100 nF
V3V3D
C35
100 nF
DUT Hdr
J5
Würth Elektronik 61302021121
VDUT
VBAT
Self test Node
R38
0 Ω
R39
0 Ω
R40
0 Ω
Hard Pull-Downs
U11
DVDD
DIN
SCLK
CS
DOUT
INT
CNVT
ADC_INT_REF
DAC_REF
D0P
D0N
AVDD
AGND
PORT0
D1P
D1N
PORT1
AVDDIO
PORT2
PORT3
PORT4
PORT5
AVSSIO
PORT6
PORT7
AGND1
AVDDIO
PORT8
PORT9
PORT10
PORT11
DGND
EP
Maxim Integrated MAX11311GTJ+
V12VA
VNEG2V5A
SPI_MOSI
SPI_SCLK
SPI_CS
SPI_MISO
INT
CVRT
R41
10 Ω
R43
10 Ω
Q2B
Q2A
Rohm UT6K3TCR
R42 DNP
R44 DNP
V5VA
C36
10 uF
50.0 V
C37
100 nF
C38
10 uF
50.0 V
REF2V5
Self test Node
R45
DNP
C39
100 nF
GND
DAC Controlled LDO
U12
OUT
IN
FB
EN
PAD
GND
DNC
Texas Instruments TLV75901PDRVR
VSYS
Variable Supply Controlled by DAC
LDO_ADJ
R47
5.1 KΩ
R46
16 KΩ
R49
5.1 KΩ
C45
2.2 µF
C46
10 nF
C47
1.00 µF
R48
10 KΩ
C40
1.00 µF
C41
100 nF
C42
1.00 µF
C43
1.00 µF
C44
100 nF
Title
TEST HEADER 1
Size
Letter
Number
a44
Revision
02
Date:
9-08-2025
File:
Hdr1.SchDoc
Sheet of
Drawn By:
Sistemi



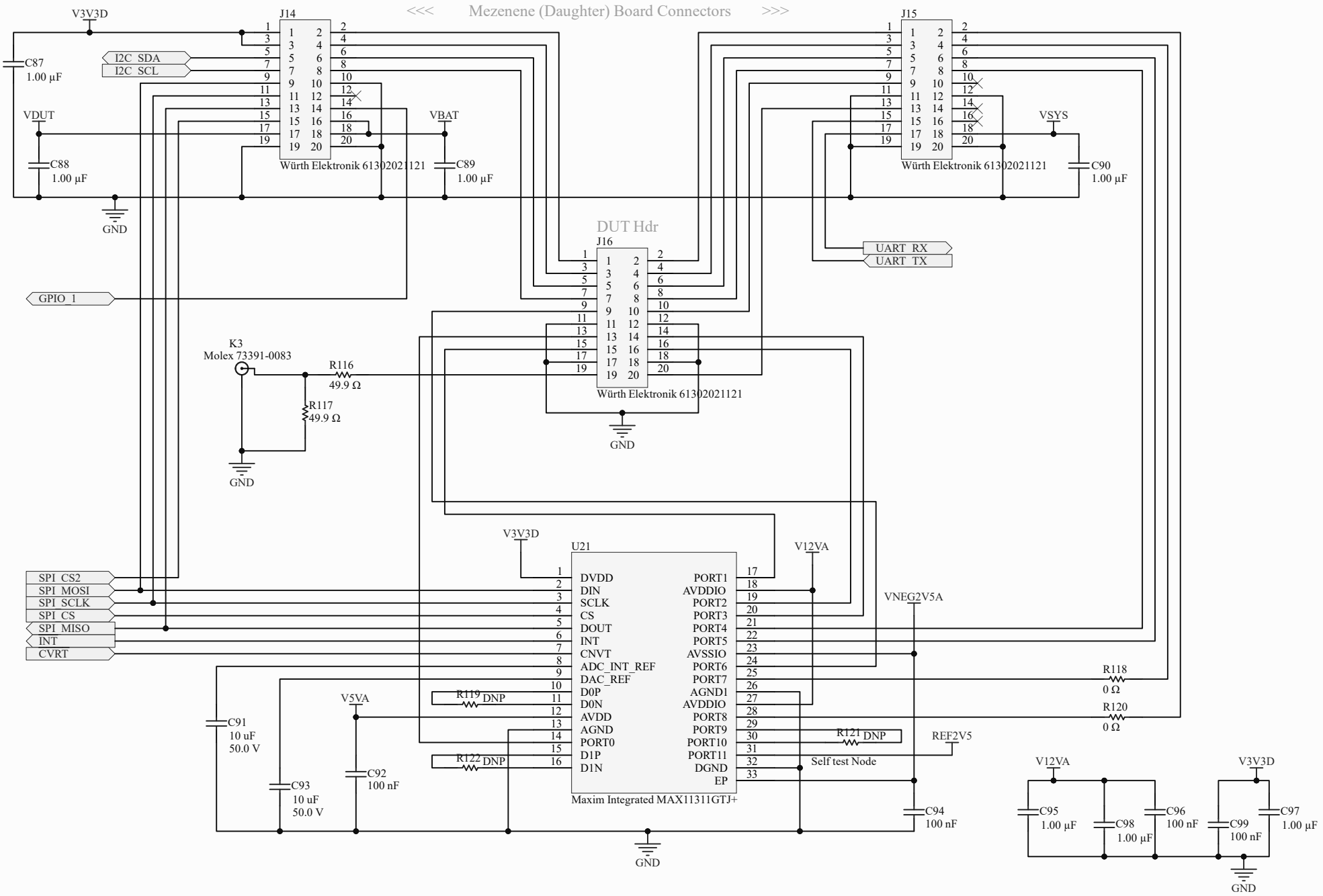
Title	TEST HEADER 2	
Size: Letter	Number: a44	Revision: 01
Date: 9-08-2025		Sheet of
File: Hdr2.SchDoc		Drawn By: Sistemi



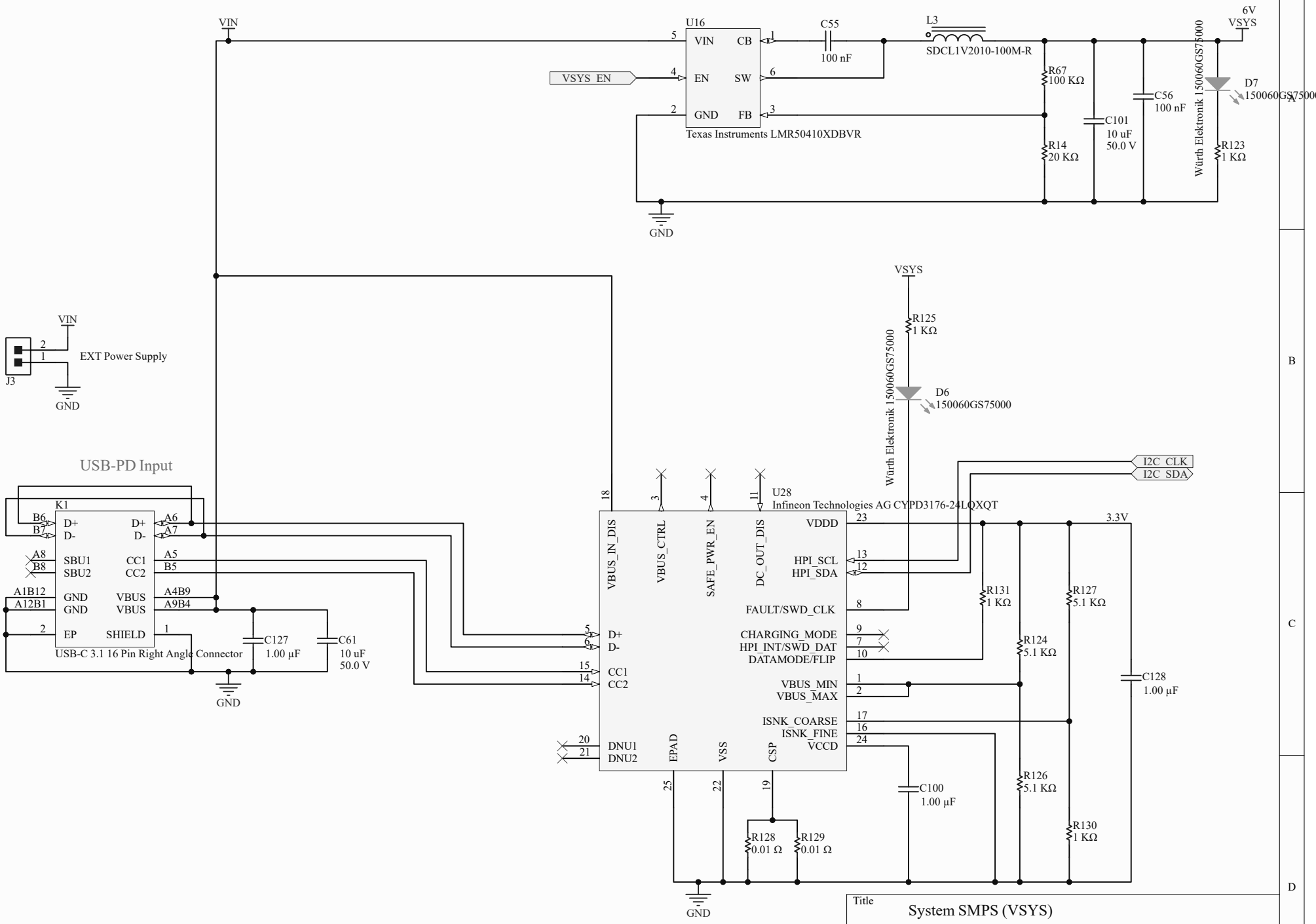
Title	TEST HEADER 3	
Size: Letter	Number: a44	Revision: 02
Date: 9-08-2025		Sheet of
File: Hdr3.SchDoc		Drawn By: Sistemi

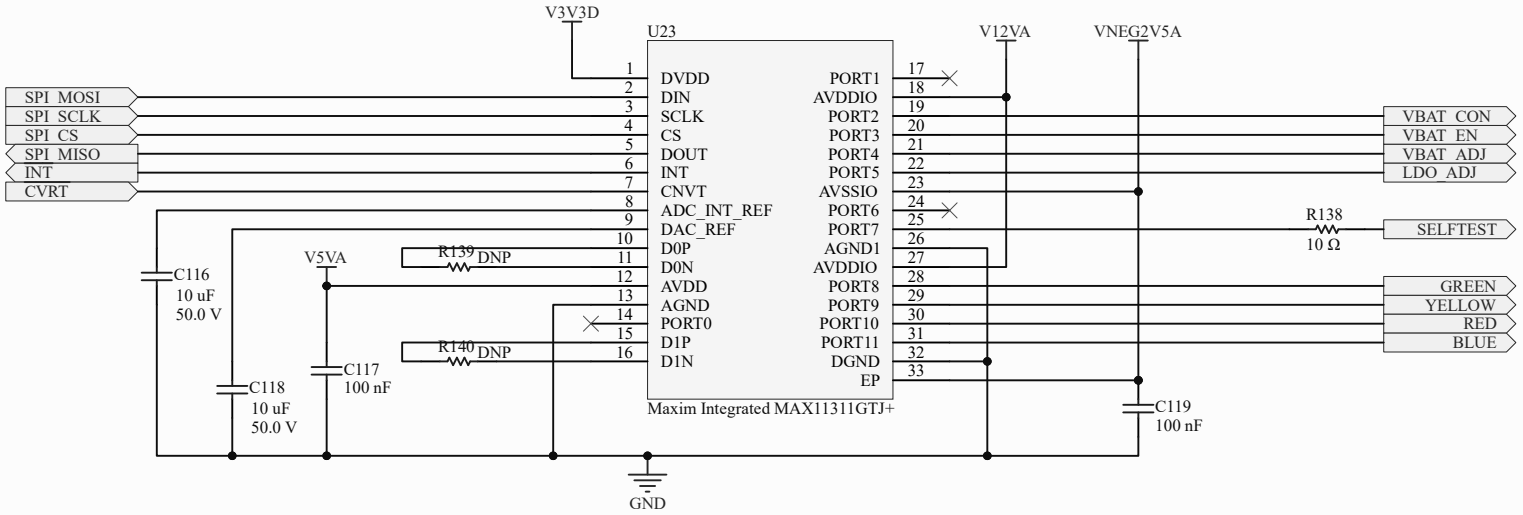


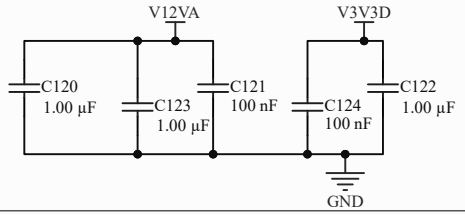
Title	STLINK-V3MODS		
Size: Letter	Number: a44		Revision: 02
Date: 9-08-2025		Sheet of	
File: stlink.SchDoc		Drawn By: Sistemi	



Title	TEST HEADER 4	
Size: Letter	Number: a44	Revision: 01
Date: 9-08-2025		Sheet of
File: Hdr4.SchDoc		Drawn By: Sistemi


VIN
U16
VIN
CB
EN
SW
GND
FB
Texas Instruments LMR50410XDBVR
C55
100 nF
L3
SDCL1V2010-100M-R
6V
VSYS
VSYS_EN
R67
100 KΩ
C56
100 nF
C101
10 uF
50.0 V
R14
20 KΩ
D7
150060GS75000
Würth Elektronik 150060GS75000
R123
1 KΩ
GND
VSYS
R125
1 KΩ
D6
150060GS75000
Würth Elektronik 150060GS75000
VIN
J3
EXT Power Supply
GND
USB-PD Input
K1
USB-C 3.1 16 Pin Right Angle Connector
C127
1.00 µF
C61
10 uF
50.0 V
GND
U28
Infineon Technologies AG CYPD3176-24LQXQT
VBUS_IN_DIS
VBUS_CTRL
SAFE_PWR_EN
DC_OUT_DIS
VDDD
HPI_SCL
HPI_SDA
FAULT/SWD_CLK
CHARGING_MODE
HPI_INT/SWD_DAT
DATAMODE/FLIP
VBUS_MIN
VBUS_MAX
ISNK_COARSE
ISNK_FINE
VCCD
D+
D-
CC1
CC2
DNU1
DNU2
EPAD
VSS
CSP
I2C_CLK
I2C_SDA
3.3V
R131
1 KΩ
R127
5.1 KΩ
R124
5.1 KΩ
C128
1.00 µF
R126
5.1 KΩ
R130
1 KΩ
C100
1.00 µF
R128
0.01 Ω
R129
0.01 Ω
GND
Title
System SMPS (VSYS)
Size
Letter
Number
a44
Revision
02
Date:
9-08-2025
Sheet of
File:
SMPS.SchDoc
Drawn By:
Sistemi





Title: IO Expand (for Teensy)		
Size: Letter	Number: a44	Revision: 02
Date: 9-08-2025		Sheet of
File: IOExpand.SchDoc		Drawn By: Sistemi